

One source of data for this lab is the public Lahman database which contains a number of data sets with different units of observation. Below are the first few rows and some of the columns for two of these data sets: Teams and Batting. They contain data going back to 1871. Use these excerpts to help you answer the following questions.

yearID	teamID	franchID	G	W	L	R	RA	name
1996	PHI	PHI	162	67	95	650	790	Philadelphia Phillies
1984	NYN	NYM	162	90	72	652	676	New York Mets
1943	CIN	CIN	155	87	67	608	543	Cincinnati Reds
1940	SLN	STL	156	84	69	747	699	St. Louis Cardinals
1922	SLN	STL	154	85	69	863	819	St. Louis Cardinals
1888	CN2	CIN	137	80	54	745	628	Cincinnati Red Stockings

playerID	yearID	teamID	G	AB	R	H	BB	SO
plunker01	1997	CLE	56	1	0	0	0	1
tackeje01	1993	BAL	39	87	8	15	13	28
maccofr01	1976	DET	9	3	0	0	0	2
demaejo01	1957	KC1	135	461	44	113	22	82
chaseke01	1940	WS1	35	92	8	15	4	22
coombja01	1908	PHA	78	220	24	56	9	34

Question 1

What is the unit of observation for the Teams data set? What about for the Batting data set?

Question 2

Write out two questions about baseball that could answered purely through *summaries* of these data sets (numerical summaries or plots).

Question 3

Write out *predictive* questions (two classification and two regression) that you could answer about baseball using the data sets above. Identify a response variable for each.

Question 4

What is a question that we would need more granular (measured on a finer/more specific part of the game) data than the Teams and Batting data sets provide to answer?

Question 5

Roughly since 1962 MLB Teams have played 162 games in a season. What do you think the distribution of wins (w) looks like? Sketch a plot of what you think the *entire* wins column looks like, adding axis tick marks with plausible values, and describe the shape in words.

Question 6

What do you think the relationship between wins (w) and runs (R) looks like? Sketch a plot, adding axis tick marks with plausible values, and describe the shape in words.

Question 7

Some people believe analytics is ruining baseball because Teams are more cautious which makes the games less entertaining. Do you agree or disagree? Why? Answer in two or more sentences.